CUNY GRADUATE CENTER

# Model-Based Analysis: Missing Data & Poisson Regression Modeling

# Missing Data Methods

- Missing data imputation is a well-developed area of applied statistics, with methods ranging from simple single-value substitutions to fully model-based approaches. The major strategies are organized below from simplest to most principled.
- Single Imputation Methods
- Mean/Median/Mode Substitution replaces missing values with the observed mean (for continuous variables), median (for skewed distributions), or mode (for categorical variables). It is easy to implement but artificially reduces variance, distorts the variable's distribution, and attenuates correlations with other variables. It is generally discouraged in serious analytical work.

- Hot Deck Imputation replaces a missing value with an observed value drawn from a "similar" respondent in the same dataset, typically defined by matching on a set of auxiliary variables. It preserves the empirical distribution better than mean substitution and has a long history in survey research, particularly at agencies like the U.S. Census Bureau.
- Cold Deck Imputation is the same logic, but the donor value comes from an external source — a prior wave of the survey, a reference dataset, or historical records — rather than the current dataset.

- Regression Imputation predicts the missing value from a regression model fitted on cases with complete data, using observed variables as predictors. It produces plausible, individually tailored imputed values but still underestimates uncertainty because it treats imputed values as known quantities with no residual error.
- Stochastic Regression Imputation adds a randomly drawn residual to each regression-predicted value, restoring some of the natural variability that deterministic regression imputation suppresses. It is a meaningful improvement over plain regression imputation and is conceptually close to the multiple imputation approach described below.

Multiple imputation, formalized by Donald Rubin in the 1980s, is the current methodological standard for handling missing data in most research contexts. Rather than filling in a single value, MI generates m complete datasets (typically 20–50 in modern practice), each with slightly different imputed values drawn from the posterior predictive distribution of the missing data given the observed data. The analyst fits the desired model on each of the m datasets separately, then combines the results using Rubin's Rules — which pool point estimates by averaging and propagate both within-imputation and between-imputation variance into the final standard errors. This two-source variance accounting is what makes MI statistically valid: it honestly reflects the uncertainty attributable to not knowing the true missing values.

- The most widely used algorithm for generating the imputed datasets is Multivariate Imputation by Chained Equations (MICE), also called fully conditional specification. MICE imputes each variable with missing data sequentially using a separate regression model conditioned on all other variables, cycling through the variables iteratively until convergence. Its flexibility — each variable can use a regression model appropriate to its type (linear, logistic, ordinal, Poisson, etc.) — has made it the dominant practical implementation of MI.

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## Summary

- For most applied research, multiple imputation via MICE or FIML represents best practice, as both are statistically valid under the MAR assumption, make efficient use of all available data, and properly propagate missing-data uncertainty into final inferences. Simpler methods such as mean substitution or complete-case analysis are generally acceptable only when missingness is minimal and demonstrably MCAR.

- A Critical Distinction: Missing Data Mechanisms
- The appropriateness of any imputation method depends heavily on the assumed missing data mechanism, as formalized by Rubin:
- Missing Completely at Random (MCAR): Missingness is unrelated to any data, observed or unobserved. Even listwise deletion is unbiased here, though inefficient.
- Missing at Random (MAR): Missingness depends on observed variables but not on the missing values themselves, after conditioning on those observed variables. MI and FIML are valid under MAR, which is the key assumption justifying their use.
- Missing Not at Random (MNAR): Missingness depends on the unobserved values themselves — for example, heavy smokers being more likely to skip questions about smoking. No standard imputation method fully corrects for MNAR; it requires sensitivity analyses or specialized selection models.

# Poisson Distribution of Cigarette Smoking